SEMIN JIN 진세민

COMPUTATIONAL & AI-DRIVEN DESIGN RESEARCHER

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Computational design researcher bridging design and AI in two directions. **Design → AI**: translating designer insight into **3D scene generation**, **anthropometric simulation**, and **human-action prediction** so machines produce environments that fit real human behaviors. **AI → Design**: bringing **LLMs**, **diffusion**, **vision-language models**, and **empirical user studies** into the creative process to support, analyze, and reshape how designers work.

EDUCATION

Sep 2024 – Aug 2026
Doctor of Philosophy in Interior Architecture Design
Hanyang University, Seoul · Graduate School of Interior Architecture Design
Advisor: Prof. Kyung Hoon Hyun | **GPA 4.40 / 4.50 · 33 credits**
Dissertation: *Anthropometric Simulation and Human Pose Evaluation for Generative Spatial Design.*

Mar 2022 – Aug 2024
Master of Science in Interior Architecture Design
Hanyang University, Seoul · Graduate School of Interior Architecture Design
Advisor: Prof. Kyung Hoon Hyun | **GPA 4.44 / 4.50 · 27 credits**
Thesis: *Data-Driven Aesthetic Assessment Systems for Supporting Interior Design.*

Mar 2018 – Feb 2022
Bachelor of Arts in Architectural Interior Design
Kaywon University of Art & Design, Uiwang
Top-up program (2021 – 2022) | **GPA 4.23 / 4.50 · 23 credits**
Preceded by Associate of Arts (A.A.) in Interior Architecture Design, 3-yr (2018 – 2021) | **GPA 4.09 / 4.50 · 122 credits**

PUBLICATIONS

Impact Factors based on 2024 JCR · h5-index from Google Scholar

JOURNAL ARTICLES

[1] **Jin, S.**, & Hyun, K. H. (2026). Human-operational 3D indoor layout generation with LLM-driven anthropometric simulation. *Automation in Construction*, 184, 106846.
SCIE · **IF 11.5** · Top 1% · First author

[2] **Jin, S.**, & Hyun, K. H. (2026). Blending designer insights with predicted crowd-based color compatibility in interior color design. *Color Research and Application*.
SCIE · **IF 1.4** · First author

[3] Seo, J., **Jin, S.**, Choi, K., Hyun, K. H., & Joung, J. (2026). An explainable AI-based approach for identifying interior design style principles. *Advanced Engineering Informatics*, 69, 103879.
SCIE · **IF 9.9** · Top 2%

[4] **Jin, S.**†, Choi, J.†, & Hyun, K. H. (2025). CMLsearch: Semantic visual search and simulation through segmented colour, material, and lighting in interior image. *Journal of Computational Design and Engineering*, 12(1), 179–299.
SCIE · **IF 6.1** · Top 7% · † Co-first author

[5] Jo, T. H., Choi, J., **Jin, S.**, Lee, S. W., Jang, Y., Park, S. W., Ban, S., & Hyun, K. H. (2026). Workflow-Net: Toward understanding designer workflows in generative AI-driven systems through comparing node- and prompt-based interfaces. *Archives of Design Research*, 39(2), 7–34.
Scopus

[6] **Jin, S.**, & Hyun, K. H. (2024). Enhancing the interior design process with crowdsourced furnishing pairing recommendations. *Archives of Design Research*, 37(2), Vol. 150.
Scopus · KSDS Best Paper 2024 · First author

CONFERENCE PROCEEDINGS

[7] **Jin, S.**†, Kim, D.†, Ryu, J., & Hyun, K. H. (2026, April). Behavior-Aware Anthropometric Scene Generation for Human-Usable 3D Layouts. In *Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI '26)*, Barcelona, Spain.
BK Top-tier Conf. · **h5-index 139** · † Co-first author

[8] Lee, S. W., Jo, T. H., **Jin, S.**, Choi, J., Yun, K., Bromberg, S., Ban, S., & Hyun, K. H. (2024). The impact of sketch-guided vs. prompt-guided 3D generative AIs on the design exploration process. In *Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI '24)*, Honolulu, HI, USA.
BK Top-tier Conf. · **h5-index 139**

PATENTS

REGISTERED — **인테리어 제품 검색 서비스 제공 방법 및 장치**
Patent No. KR 10-2943193, Registered March 19, 2026 · 현경훈, **진세민**, 최지인

FILED — **인체측정 데이터 기반 3차원 실내 공간 레이아웃 생성 및 사용성 검증 시스템 및 방법**
Application No. KR 10-2026-0040371 · 현경훈, **진세민**, 김동혁

FILED — **다중 에이전트를 기반으로 창의적 아이디어를 구조화하고 시각화하는 시스템 및 방법**
Application No. KR 10-2026-0036677 · 현경훈, **진세민**, 이승원

FILED — **디자인 아티팩트 기반 이미지 생성 시스템 및 방법**
Application No. KR 10-2025-0184966 · 현경훈, 이승원, 최지인, **진세민**

FILED — **군중 선호도 데이터를 활용한 공간 디자인 이미지의 설명적 평가 및 색채 조합 생성 시스템과 방법**
Application No. KR 10-2024-0054887 · 현경훈, **진세민**

DESIGN · REGISTERED — **화면디자인이 표시된 디스플레이 패널**
Registered Design · Application No. KR 30-2026-0004673 · 현경훈, **진세민**, 이승원

EXPERIENCE

Spring 2026 — **Teaching Fellow — 4-week module**
Department of Interior Architecture Design, Hanyang University
Course: *Furniture Design and Digital Fabrication* (English-medium)

Spring 2025 — **Teaching Fellow — 4-week module**
Department of Interior Architecture Design, Hanyang University
Course: *Furniture Design and Digital Fabrication* (English-medium)

2022 – Present **Graduate Researcher**
Design Informatics Lab · Human-Centered AI Design Institute, Hanyang University

AWARDS, CERTIFICATIONS & SERVICE

AWARDS

Nov 2024 **Best Paper Award (Excellence)**
Korean Society of Design Science (KSDS) · 2024 Annual Conference
For "Enhancing the Interior Design Process with Crowdsourced Furnishing Pairing Recommendations," *Archives of Design Research*, Vol. 37, No. 2.

Nov 2022 **Excellence Award · 우수작품상**
Korean Institute of Interior Design (KIID) · 30th Anniversary International Invitational Exhibition
Work: *Return to Flow of Nature.*

CERTIFICATIONS

2021 **Engineer Colorist, 컬러리스트기사**
Korea National Technical Qualification · Human Resources Development Service of Korea (HRDK)

ACADEMIC SERVICE

Apr 2026 **Workshop Paper — Beyond Semantic Similarity: Open Challenges for Embedding-Based Creative Process Analysis Across AI Design Tools**
Herding Cats Workshop · ACM CHI 2026, Barcelona, Spain

Oct 2025 **Demo Presentation**
Mind & Motion Workshop · ACM UIST 2025, Busan, Korea

2024 – Present **Reviewer**
ACM CHI · ACM UIST · ACM Creativity & Cognition (C&C)